

RESEARCH INTERESTS

I am broadly interested in Computational Linguistics and Natural Language Processing, specifically (1) Human AI alignment, (2) Developing robust evaluation frameworks for LLMs, and (3) Investigating model internals to advance LLM interpretability and AI safety.

EDUCATION

Jahangirnagar University

Master of Science in Computer Science

Dhaka, Bangladesh

May 2025 – May 2026

- **Current CGPA:** 3.97/4.0
- **Relevant Coursework:** Machine Learning (A+), Deep Learning (A+), Natural Language Processing (A+), Data Mining (A), Artificial Intelligence & Expert Systems (A+), Data Structures & Algorithms (A+), Software Engineering & Database Systems (A+), Computer Architecture & Operating Systems (A+), Programming Languages (A+)

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Civil Engineering, with Honors

Dhaka, Bangladesh

Feb. 2020 – Mar. 2025

- **CGPA:** 3.77/4.0, **Major:** Transportation Engineering, **Minor:** Environmental Engineering
- **Relevant Coursework:** Differential Equations & Statistics (A+), Computer Programming Sessional (A+), Engineering Computation Sessional (A+), Differential, Integral Calculus and Matrices (S) [†]

[†]Satisfactory; University graded S/F in 2020-21 due to COVID-19.

RESEARCH EXPERIENCE

Graduate Student Researcher

JU CIDL Research Lab

Aug. 2025 – Present

Supervisor: Dr. Md Musfique Anwar

- **Beyond Clean Text: Evaluating Encoder and Decoder Robustness for Bangla Event Detection in Noisy Text** [\[arXiv\]](#)
Tanvir Ahmed Sijan, S. M Golam Rifat, Nayeemul Islam, Md. Musfique Anwar
 - * Developed an annotated Bangla event detection dataset and evaluation framework to assess the performance of instruction-tuned LLMs and encoder-based models on noisy, ASR-transcribed news text.
 - * Created perturbed datasets to investigate model robustness and characterize performance degradation under varying levels of Word Corruption Rate.
- **BanglaSocialBench: A Benchmark for Evaluating Sociopragmatic and Cultural Alignment of LLMs in Bangladeshi Social Interaction**
Tanvir Ahmed Sijan, S. M Golam Rifat, Pankaj Chowdhury Partha, Md. Tanjeed Islam, Md. Musfique Anwar
*ACL Student Research Workshop 2026 (Oral) · **Best Paper Award** · [\[ACL Anthology\]](#)*
 - * Conceptualized and led the development of BanglaSocialBench, an annotated benchmark comprising 1,719 culturally grounded instances across three domains for evaluating the sociopragmatic competence of large language models.
 - * Benchmarked 12 LLMs in zero-shot setting, uncovering over-politeness, religious kinship term conflation, and rigidity in ambiguous social contexts.

Undergraduate Thesis

Bangladesh University of Engineering and Technology (BUET)

Nov. 2023 – Feb. 2025

Supervisor: Dr. Md. Shamsul Hoque

- **Assessing the Visual and Built Environment Risk on Pedestrian-Vehicle Crash Severity in Absence of Detailed Roadway Inventory: A Deep Learning Based Framework Harnessing Public Geospatial Data**
Tanvir Ahmed Sijan, Md Asif Raihan, Sk. Md. Mashrur, Md. Shamsul Hoque
Submitted to Accident Analysis & Prevention, 2025
 - * Built an end-to-end multimodal data pipeline integrating Google Street View API, Overture Maps, OpenStreetMap, and crash datasets to generate urban risk features for pedestrian crash severity prediction.

- * Applied semantic segmentation (Detectron2 + Mask2Former) on 12K+ street-view images and trained ensemble ML models (XGBoost, Random Forest, AdaBoost, CatBoost) with SHAP-based interpretability to identify key environmental risk factors.

WORK EXPERIENCE

Maks Inc.

Dhaka, Bangladesh (HQ: Frisco, TX, USA)

Research Intern

May 2025 – Sep 2025

Projects: Study on Heavy-Duty Vehicles (**World Bank Group**), Akhaura–Tamabil Land Port Feasibility Study (**Asian Development Bank**)

- Analyzed large-scale BRTA fleet data to model vehicular emissions, performed spatial analysis with QGIS, conducted nationwide stakeholder interviews, and synthesized findings for policy-oriented technical reports.

HONORS & AWARDS

Best Paper Award

Jul. 2026

ACL Student Research Workshop 2026

Undergraduate Student Research Grant

Jan. 2024

Research and Innovation Centre for Science and Engineering, BUET

Dean's List Scholarship

Jul. 2019

Bangladesh University of Engineering and Technology (BUET)

- Awarded for maintaining an average GPA of 3.75 or above across two consecutive regular terms.

3rd Place, Concrete Solutions Competition

Mar. 2022

American Concrete Institute (ACI)

- Recognized for the most innovative team design; project featured in the **Daily Prothom Alo** [\[Feature\]](#)

Higher Secondary Certificate (HSC) Scholarship

2019

Board of Intermediate and Secondary Education, Rajshahi

- Awarded **Talentpool Scholarship**; ranked **10th (top 0.01%)** among all candidates in the Rajshahi Board.

LEADERSHIP & VOLUNTARY WORKS

ITE BUET Student Chapter

Dec. 2024 – Mar. 2025

Deputy Vice President

- Collaborated with other executive members to organize workshops and seminars on various topics of transportation research.

ACI BUET Student Chapter

Mar. 2024 – Mar. 2025

Vice President

- Organized poster competitions, seminars, and field visits in coordination with chapter members.

SKILLS

- Programming:** Python, C, C++, SQL, MATLAB, LaTeX
- Libraries:** NumPy, PyTorch, Transformers, Matplotlib, Scikit-Learn, Selenium
- Other:** Git, LaTeX, Bash
- Languages:** Bangla (Native), English, Hindi (Basic)